

ASSESSMENT AND INTERNAL VERIFICATION FRONT SHEET (Individual Criteria)

(Note : This version is to be used for an assignment brief issued to students via Classter)

Course Title	IT6-A4-23 BSc Year 3 - Digital Games Development			Lecturer Name & Surname	Owen Sacco	
Unit Number & Title		ITDGM-606-2309 - Game AI				
Assignment Number, Title / Type		Game AI Home Assignment				
Date Set		19/12/2025	Deadline Date	28/01/2026		
Student Name			ID Number		Class / Group	

Assessment Criteria	Maximum Mark
A&A1 <i>Implement functional FSMs or Behaviour Trees to control character behaviours in a game environment.</i>	7
A&A2 <i>Configure NavMesh-based system for agents to perform pathfinding efficiently.</i>	7
A&A3 <i>Develop a basic Utility AI system that enables an agent to prioritise actions based on multiple factors.</i>	7
E&C1 <i>Design a hierarchical decision-making system capable of managing multiple agent goals effectively.</i>	10
E&C2 <i>Develop a reward function or curriculum learning strategy that significantly improves agent learning efficiency or performance.</i>	10
R&U6 <i>Describe various techniques for optimising AI performance and debugging AI systems in a game engine.</i>	5
R&U7 <i>Explain key ethical considerations and potential biases related to AI development in interactive entertainment.</i>	5
A&A5 <i>Implement a multi-agent system demonstrating either cooperative or competitive behaviour, including adversarial self-play.</i>	7
E&C3 <i>Integrate various distinct AI techniques together</i>	10
Total Mark	68

Notes to Students:

- This assignment brief has been approved and released by the Internal Verifier through Classter.
- Assessment marks and feedback by the lecturer will be available online via Classter (<http://mcast.classter.com>) following release by the Internal Verifier
- Students submitting their assignment on Moodle/VLE will be requested to confirm online the following statements:

Student's declaration prior to handing-in of assignment

❖ I certify that the work submitted for this assignment is my own and that I have read and understood the respective Plagiarism Policy

Student's declaration on assessment special arrangements

❖ I certify that adequate support was given to me during the assignment through the Institute and/or the Inclusive Education Unit.

❖ I declare that I refused the special support offered by the Institute.



Assignment: Game AI

Home Assignment

Assignment Guidelines - Read the following instructions carefully:

- The assignment coversheet should be the first sheet in your assignment. Moreover, the coversheet should be fully completed with all the necessary details.
- You are required to answer **ALL** questions.
- You are required to use Unity and C# for this assignment.
- All text/code must be properly referenced. In the absence of proper referencing, the assignment will be regarded as plagiarised.
- **Copying is strictly prohibited and will be penalised** in line with the College's disciplinary procedures.
- The deadline to submit all deliverables is **28/01/2026**
- You need to submit all your assignment deliverables (as explained in the assignment brief) via the links on Moodle/VLE.
- You need to commit all your C# scripts and project files to a Git repository. It is important to commit your code in stages (**minimum 20 commits**) to show your progress whilst working on this assignment. Include clear comments and notes to explain key concepts in your code.
- You are required to record a (video) demonstration of all your deliverables and upload your recording to any streaming service of your choice. Ensure that you set the correct permissions and share it only with your lecturer. Kindly limit the recording to not more than 20 minutes.

Assignment Description

You are required to develop a Unity project featuring a **Battle Arena** scenario. This project is designed to integrate and demonstrate your understanding of three distinct AI techniques simultaneously: traditional navigation, advanced decision-making, and reinforcement learning.

You must also submit a recorded video presentation (not more than 20 minutes) to demonstrate the project, explain your architecture and technical decisions, and discuss ethical considerations.

Game Design

The *Battle Arena* scenario is a three-way skirmish where three distinct autonomous agents compete to survive and eliminate the others. The final running scene must contain one instance of each of the three agent types (i.e. Guard, Strategist, Gladiator). No human player is required.

The Arena Environment: The scene should be an enclosed environment containing:

- Static obstacles (for example walls) that require pathfinding.
- Critical resources (for example *Health Packs*, *Ammo Crates*) that the Strategist (Task 2) must track and prioritise.
- A clear line-of-sight setup to allow agents to detect each other.

NPC Roles and Objectives (the NPCs will compete/fight against each other):

1. *The Guard* (FSM/Behaviour Tree Agent): Patrol the environment and pursue any hostile agent (Strategist or Gladiator) that enters its vision. It is the most predictable agent, relying purely on its FSM states or Behaviour Tree and pathfinding rules.

2. *The Strategist* (Utility AI Agent): Survive by prioritising its actions based on its current utility score. It will actively seek resources and hide/flee when critical, only engaging in direct combat when necessary. It is the most adaptable agent.
3. *The Gladiator* (ML Agent): Survive and win by eliminating the others. Its behaviour is learned entirely from the reward signals, resulting in complex, non-deterministic combat behaviours.

All three agents must be able to detect, track, and attempt to eliminate the other two, providing a clear test bed for the integration task.

Practical Tasks [58 marks]

The project is broken down into four development tasks that should all be integrated together within the whole project.

Task 1: The Guard (FSM & Navigation/Pathfinding) [14 Marks]

The Guard will be a traditional, rule-based NPC, controlled by a pathfinding system and a Finite State Machine (FSM) or Behaviour Tree.

- **1.1 Pathfinding Implementation (A&A2) [7 marks]:** You must implement **ONE** of the following pathfinding methods for the Guard. This task focuses on configuring the environment and utilising the system to move the agent efficiently.
 - o **Option A: NavMesh:** Configure a **Unity NavMesh** for the arena. The Guard agent must be able to successfully navigate efficiently around both static (for example walls) and dynamic (for example moving objects) obstacles.
 - o **Option B: Grid-based A* Pathfinding:** Utilise the **A* Pathfinding Project** to manage navigation. The agent must be able to find the shortest path between two points while avoiding obstacles defined by the grid (for example walls, pits).

- **1.2 FSM or Behaviour Tree Implementation (A&A1) [7 marks]:** Script a **Finite State Machine (FSM)** or use a **Behaviour Tree** to control the Guard's logic. The FSM/Behaviour Tree must handle **at least three distinct states** - *for example: Patrol* (following a path), *Chase* (pursuing a target), and *Search* (investigating a last known location). Show that the Guard's movement, speed, and decision-making logic change significantly and appropriately based on its current state.

Task 2: The Strategist (Advanced Decision Making) [17 Marks]

The Strategist will use a decision-making system capable of weighing up multiple factors to choose the best action based on current needs.

- **2.1 Utility AI System & Scoring (A&A3) [7 marks]:** Develop a **Utility AI system** to drive the Strategist's actions (for example *Heal*, *Collect Ammo*, *Hide*). The system must actively calculate a score for multiple potential actions at every decision interval. Actions must be scored based on dynamic factors such as health percentage, distance to threat, or current resources. The Strategist should never perform the same sequence of actions twice and should appear to make strategic, context-aware decisions.
- **2.2 Hierarchical Goal Management (E&C1) [10 marks]:** Implement a **hierarchical approach or Goal-Oriented Action Planning (GOAP)** that allows a critical goal like *Survival* to immediately override the standard Utility AI output when a critical condition is met (for example low health). Demonstrate effective conflict resolution between competing high-level goals.

Task 3: The Gladiator (Machine Learning) [17 Marks]

The Gladiator will be a reinforcement learning agent trained using Unity's ML-Agents toolkit, relying only on reward signals for learning.

- **3.1 Training Strategy (E&C2) [10 marks]:** Using Unity ML-Agents, implement a *reinforcement learning* approach, and utilise a *reward shaping* strategy to train an

agent to survive and compete effectively. The submitted agent model must exhibit complex, non-scripted, emergent behaviour.

- **3.2 Multi-Agent Scenario (A&A5) [7 marks]:** Implement a **Multi-Agent scenario** during the training phase. This is achieved by running multiple instances of the *Gladiator* agent simultaneously, typically structured as adversarial self-play. The goal of this task is to produce a single, highly-skilled policy/model file (.nn or .onnx) that has been trained via competition with other instances of itself. This single trained agent will be the one used in the final integration scene (Task 4). The trained .nn model file must be submitted alongside the code.

Task 4: Integration (Architecture) [10 Marks]

This task combines all the individual AI systems into a single functional environment.

- **4.1 Scene Integration (E&C3) [10 marks]:** Combine all three distinct agent types (one *Guard*, one *Strategist*, one *Gladiator*) into a single, functional game scene. Ensure all distinct AI paradigms operate simultaneously and interact logically without conflicts or errors. Demonstrate that the three different AI architectures can coexist and compete within the same game loop.

Task 5: Video Presentation & Analysis [10 Marks]

Record and submit a video recording of not more than 20 minutes. **You must narrate the video yourself.**

- **5.1 Code Walkthrough:** Briefly demonstrate and verbally explain the logic of your code implementations.
- **5.2 Optimisation (R&U6) [5 Marks]:** In your recording, show also the Unity Profiler running while your scene is active. Verbally describe the techniques you used to identify bottlenecks and optimise the AI systems (for example Object Pooling, Component Caching etc).

- **5.3 Ethics (R&U7) [5 Marks]:** Verbally explain the key ethical considerations in Game AI. Discuss the risk of potential biases and propose strategies developers can use to ensure fairness. Demonstrate critical thinking about the societal and player impact of AI design.

Submission Requirements

Please ensure all files are correctly packaged and submitted before the deadline.

1. **Unity Project Build:** A playable executable of your project.
2. **Source Code:** All code, configuration files and trained models should be committed to a Git repository. Commit your files in stages (minimum 20 commits) to show your progress whilst working on this assignment. Include clear comments when committing.
3. **Video File:** An .mp4 file or a working link to a private/unlisted video upload (YouTube/OneDrive).

Grading Scheme

Task	Assessment Criteria	Marks Awarded	Task Marks
1	1.1 Pathfinding Implementation (A&A2) [7 marks] 1.2 FSM or Behaviour Tree Implementation (A&A1) [7 marks]		14
2	2.1 Utility AI System & Scoring (A&A3) [7 marks] 2.2 Hierarchical Goal Management (E&C1) [10 marks]		17
3	3.1 Training Strategy (E&C2) [10 marks] 3.2 Multi-Agent Scenario (A&A5) [7 marks]		17
4	4.1 Scene Integration (E&C3) [10 marks]		10
5	5.2 Optimisation (R&U6) [5 Marks] 5.3 Ethics (R&U7) [5 Marks]		10
Total			68